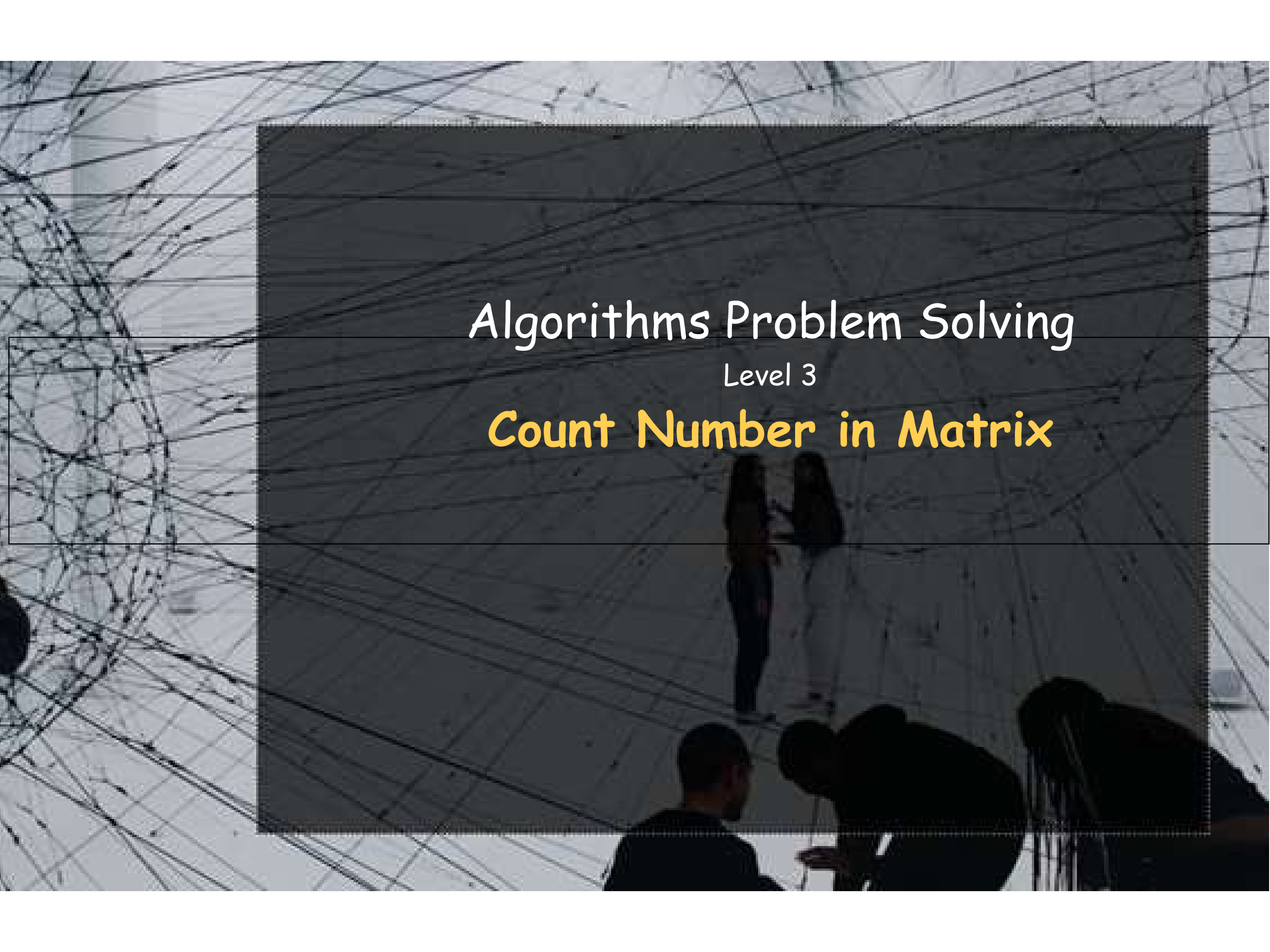


Algorithms Problem Solving

Level 3

Count Number in Matrix

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the lower part of the image, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys, blues, and the warm tones of the light coming through the glass.

Algorithms Problem Solving

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Definition

Target Frequency Counter in 2D Array

-  Program calculates how many times a specified number appears inside a matrix.

-  The result represents the frequency of that value within the entire matrix.

Explanation

Complete Matrix Traversal

-  Nested loops visit every row and column position.

-  No cell can be skipped because the goal is to calculate the exact frequency.

Target Matching

-  Each element is compared with the requested number.

-  When `matrix[i][j] == number`, the counter is incremented.

Accumulator

-  The counter starts at `0`.

-  Every matching occurrence adds `1` to the final count.

Procedures

1. Define `PrintMatrix()`.
2. Define `CountNumberInMatrix(matrix, rows, cols, number)`.
3. Read the target number from the user.
4. Traverse the complete matrix.
5. Increment the counter for every match.
6. Return and display the final count.

💡 Example

matrix1 = { {9,1,12}, {0,9,1}, {0,9,9} };

➡ Result

Number 9 count in matrix is 4

Note

- ◆ The method works with positive numbers, negative numbers, and zero.
- ◆ Exact counting requires visiting every element.
- ◆ Time complexity is $O(N \times M)$ for an $N \times M$ matrix.

⊘ Warning

- ◆ The counter must be initialized to `0`.
- ◆ Do not return immediately after finding one match because the remaining occurrences still need to be counted.
- ◆ For floating-point values, direct equality may be unreliable because of precision.



Comparison

▣ `CountNumberInMatrix()`

- ✓ Searches the entire matrix.
- ✓ Counts every matching occurrence.
- ✓ Returns an integer frequency.

▤ `IsNumberInMatrix()`

- ✓ Stops after finding the first match.
- ✓ Returns only `true` or `false`.
- ✓ Is more appropriate when only existence matters.

✦ Summary

- ✓ Traverses every cell in a 2D matrix.
- ✓ Compares each element with a target number.
- ✓ Maintains an occurrence counter.
- ✓ Returns the total frequency of the target value.

Thank you 😊